WORKSHEET 3.5

Choosing the Right Rule

Unit 3: Composite, Implicit and Inverse Functions

Key ideas

- Ask: what is the **last operation** performed when evaluating the expression? That decides the first rule.
- A product of two functions: product rule. One function inside another: chain rule. Often both, one inside the other.
- A quotient can be rewritten as a product with a negative power when that is easier.
- Simplify before differentiating when the algebra is cheap; do not simplify afterwards unless asked.

Worked examples

Example 1 · Product outside, chain inside

Find $\dfrac{d}{dx}\left[x^2 e^{3x}\right]$.

Solution. $2xe^{3x} + x^2 \cdot 3e^{3x}$.

$e^{3x}\left(2x + 3x^2\right)$.

Example 2 · Chain outside, product inside

Find $\dfrac{d}{dx}[\sin(x\cos x)]$.

Solution. $\cos(x\cos x)$ times the derivative of $x\cos x$.

$\cos(x\cos x)\left(\cos x - x\sin x\right)$.

Example 3 · Quotient with a chain

Find $\dfrac{d}{dx}\left[\dfrac{\ln(x^2+1)}{x}\right]$.

Solution. $\dfrac{x\cdot\frac{2x}{x^2+1} - \ln(x^2+1)}{x^2}$.

$\dfrac{2}{x^2+1} - \dfrac{\ln(x^2+1)}{x^2}$.

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Practice

CORE

1. Which rule is applied first to differentiate $y = (3x + 1)^4 \sin x$?

2. Which rule is applied first to differentiate $y = \sin\big((3x + 1)^4\big)$?

3. Find $\dfrac{d}{dx}\left[e^x \ln x\right]$.

4. Find $\dfrac{d}{dx}\left[(1 + \cos x)^2\right]$.

EXAM LEVEL

5. Find $\dfrac{d}{dx}\left[\sqrt{x}\, e^{-x}\right]$.

6. Find $\dfrac{d}{dx}\left[\tan^2(3x)\right]$.

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7. Find $\dfrac{d}{dx}\left[\dfrac{x}{\sqrt{x^2+1}}\right]$ in simplified form.

8. Find $\dfrac{d}{dx}[\ln(\sec x + \tan x)]$ in simplified form.

Question 9 · Free-response practice

Let $f(x) = x\sqrt{8-x^2}$ for $-2\sqrt{2} \le x \le 2\sqrt{2}$.

(a) Show that $f'(x) = \dfrac{8-2x^2}{\sqrt{8-x^2}}$.

(b) Find the values of x at which the graph of f has a horizontal tangent.

(c) Write an equation of the tangent line at the point found in part (b) with $x > 0$.

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Worked solutions

1. The product rule: the last operation is the multiplication.

2. The chain rule: the last operation is the sine.

3. $e^x \ln x + \dfrac{e^x}{x}$.

4. $2(1+\cos x)(-\sin x) = -2\sin x\,(1+\cos x)$.

5. $\dfrac{e^{-x}}{2\sqrt{x}} - \sqrt{x}\,e^{-x}$.

6. $2\tan(3x)\cdot\sec^2(3x)\cdot 3 = 6\tan(3x)\sec^2(3x)$.

7. $\dfrac{\sqrt{x^2+1} - \frac{x^2}{\sqrt{x^2+1}}}{x^2+1} = \dfrac{1}{(x^2+1)^{3/2}}$.

8. $\dfrac{\sec x\tan x + \sec^2 x}{\sec x + \tan x} = \sec x$.

9. (a) $\sqrt{8-x^2} + x\cdot\dfrac{-x}{\sqrt{8-x^2}} = \dfrac{(8-x^2)-x^2}{\sqrt{8-x^2}}$.

   (b) $8-2x^2=0$ gives $x=2$ and $x=-2$.

   (c) $f(2) = 2\sqrt{4} = 4$ and the slope is 0: the line $y=4$.

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